NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH POUND

MIL-PRF-83504/13(USAF)
18 December 1990

PERFORMANCE SPECIFICATION SHEET

SWITCHES, DUAL IN-LINE PACKAGE (DIP),
ROTARY ACTUATED, BINARY CODED, SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-83504.

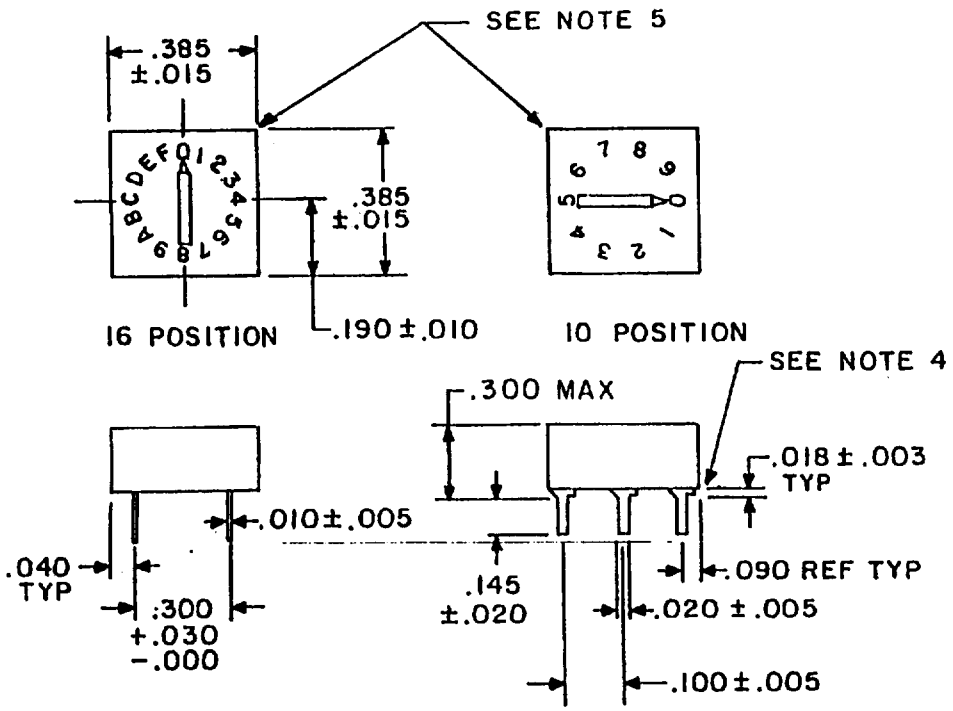



FIGURE 1. Dimensions and configuration .


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

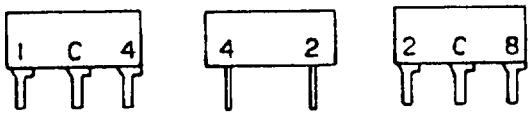
| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | 0.08 | .040 | 1.02 |
| .005 | 0.13 | .090 | 2.29 |
| .010 | 0.25 | .100 | 2.54 |
| .015 | 0.38 | .145 | 3.68 |
| .018 | 0.46 | .190 | 4.83 |
| .020 | 0.51 | .300 | 7.62 |
| .030 | 0.76 | .385 | 9.78 |

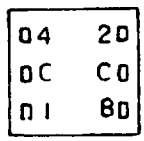
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Exact shape of switch is optional provided outline dimensions are not exceeded.
4. Design of terminals and standoffs are optional. A minimum clearance of .010 inch on the terminal above the standoff shall be free of epoxy or other sealants.
5. The housing shall be marked with position numbers as shown. Method used to indicate switch position is optional.

FIGURE 1. Dimensions and configuration - Continued.



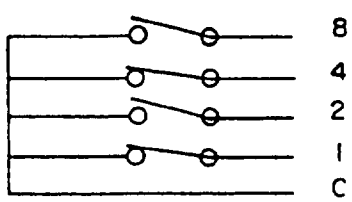
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.




| Switch position | Circuit configuration number 1 | | | |
|---|---|---|---|---|
| | Common C connection to | | | |
| | 1 | 2 | 4 | 8 |
| 0 | | | | |
| 1 | X | | | |
| 2 | | X | | |
| 3 | X | X | | |
| 4 | | | X | |
| 5 | X | | X | |
| 6 | | X | X | |
| 7 | X | X | X | |
| 8 | | | | X |
| 9 | X | | | X |

| Switch position | Circuit configuration number 2 | | | |
|---|---|---|---|---|
| | Common C connection to | | | |
| | 1 | 2 | 4 | 8 |
| 0 | | | | |
| 1 | X | | | |
| 2 | | X | | |
| 3 | X | X | | |
| 4 | | | X | |
| 5 | X | | X | |
| 6 | | X | X | |
| 7 | X | X | X | |
| 8 | | | | X |
| 9 | X | | | X |
| A | | X | | X |
| B | X | X | | X |
| C | | | X | X |
| D | X | | X | X |
| E | | X | X | X |
| F | X | X | X | X |

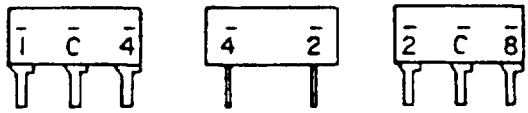


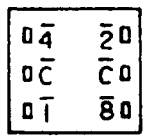

Circuit with switch in position number 5

FIGURE 2. Circuit configuration and terminal location.



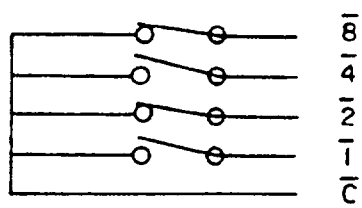
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.




| Switch position | Circuit configuration number 3 | | | |
|---|---|---|---|---|
| | Common C connection to | | | |
| | 1 | 2 | 4 | 8 |
| 0 | X | X | X | X |
| 1 | | X | X | X |
| 2 | X | | X | X |
| 3 | | | X | X |
| 4 | X | X | | X |
| 5 | | X | | X |
| 6 | X | | | X |
| 7 | | | | X |
| 8 | X | X | X | |
| 9 | | X | X | |

| Switch position | Circuit configuration number 4 | | | |
|---|---|---|---|---|
| | Common C connection to | | | |
| | 1 | 2 | 4 | 8 |
| 0 | X | X | X | X |
| 1 | | X | X | X |
| 2 | X | | X | X |
| 3 | | | X | X |
| 4 | X | X | | X |
| 5 | | X | | X |
| 6 | X | | | X |
| 7 | | | | X |
| 8 | X | X | X | |
| 9 | | X | X | |
| A | X | | X | |
| B | | | X | |
| C | X | X | | |
| D | | X | | |
| E | X | | | |
| F | | | | |




Circuit with switch in position number 5

FIGURE 2. Circuit configuration and terminal location - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Material:

Body and actuator: Molded plastic.

Terminal finish: Solderable.

Terminals: The terminals shall be free of epoxy or other sealants for their entire length to .01 inch (0.3 mm) minimum above standoff.

Dimensions and configuration: See figures 1 and 2.

Design and construction:

Base and actuator: Process sealed.

Tape cover: Integral actuator o-ring seal may be provided in lieu of tape cover.

Circuit configuration and terminal location: See figure 2.

Switch position and setting identification: See figure 1.

Orientation: See figures 1 and 2.

Operating torque: 1.0 to 9.0 inch-ounces.

Cycle of operation: A cycle shall consist of a rotation of the shaft from position number 0 to position number 9 or F and return.

Electrical ratings:

10 mA maximum at 50 mV, switching (resistive load).
75 mA maximum at 5 V dc, switching (resistive load).
15 mA maximum at 24 V dc, switching (resistive load).

Marking:

Switch housing shall be marked as shown (see figure 1).
Location of terminal marking is optional (see figure 2).

Application note: In applications where the switch will mate with sockets or with panel boards, the user should consider the compatibility of the mating terminals with respect to finishes and mechanical configuration.

Part or Identifying Number (PIN): M83504/13- (dash number from table I).

TABLE I. Dash numbers and characteristics.

| Dash number | Number of positions (continuous rotation) | Circuit configuration number |
|---|---|---|
| 001 | 10 | 1 |
| 002 | 16 | 2 |
| 003 | 10 | 3 |
| 004 | 16 | 4 |

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activity:
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(project 5930-F682)



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.